

by assuming the role of the maleficent entity. It should go beyond just the technology aspect and work on the entire chain that involves the AI system. Doing this can make its efforts more effective as it can then ensure that upstream, as well as downstream processes and systems, are tested.

A red team should consider the full ecosystem and figure out how a determined threat-actor might break it. Instead of just working towards breaking a Web app or particular technology application, it should combine several attack vectors. These attack vectors could be outside the technology domain, such as social engineering or physical access, if needed. This is necessary because although the ultimate goal would be to reduce the AI system's risks, these risks can come from many places and in many forms.

To maximise a red team's value, you should consider a scenario and goal-based exercise.

The red team should get into motion as soon the primary machine training is complete, which applies if you are developing the model inhouse. If you are outsourcing the trained model(s), then the red team must be activated as soon as sourcing completes.

The primary goal of a red team is to produce or create a plausible scenario in which the current AI system behaviour would be unacceptable—and if possible, catastrophically unacceptable. If the red team succeeds, then you can feedback its scenarios to the machine training team for retraining the model. However, if the red team does not succeed, then you can be reasonably confident that the trained model will behave reliably in the real-world scenario, too.

Carefully staging potentially-problematic scenarios and exposing the whole AI system to those situations should be one of the critical objectives. Also, this activity need not be entirely digital in format. The red team can generate these scenarios by any means available and in any formats as it seems plausible in real-life situations.

One of the ways the red team can attempt to fail the AI system is

by giving garbage inputs in primary or feedback loop and seeing how it responds. If the system is smart, it will detect the garbage and act accordingly. However, if the system magnifies or operates on the garbage input, you will know that you have work to do. These (garbage) inputs can take the form of training inputs for machine retraining.

The red team can also work on creating and providing synthetic inputs and see how the system responds. It can then use the output to examine the AI system's internal mechanics. Based on further understanding, synthetic data could be made more authentic to test the system's limits, responses and overall behaviour. Once you identify failure situations, these are easier to fix.

The red team may not necessarily try to break the system. Sometimes, it may merely cause a drift in the continuous learning of the system by feeding wrong inputs or modifying parameters and thereby causing it to fail much later.

A point where the AI system is taking input from another software or human could be a weak link. A point where output of the AI system forms an input to another API or ERP system could also be a weak link. By nature, these links are highly-vulnerable spots and weak links of the whole value-chain.

The red team should target and identify all such weak links in the system. These weak links may exist between two software systems or at the boundary of software-hardware or software-human interactions.

A red team is not for testing defences

The core objective of the red team is not to test defense capabilities. It is to do anything and everything to break the functional AI system, in as many ways as possible and by thinking outside the box. Ultimately, this should strengthen the whole organisation in the process.

Having this broader remit can enable the red team to follow intuitive methodologies for developing a reliable and ongoing learning system for the organisation. It is a promising approach to many severe problems in

the AI system's control.

However, remember that red-teaming is not equivalent to a testing team that works on generating test cases. Test cases usually follow well-defined failure condition(s), whereas for the red team the objective is much broad, and methods are undefined and often limitless.

In a nutshell, the red team should evaluate the AI system on three key parameters:

- Severity of consequences of a failure vector
- Probability of occurrence as found
- Likelihood of early detection of failure

The red team is functional, what next?

A functional red team is not just about finding holes in the AI system; it is also about providing complete guidance and playbook to improve those weak points, plug those holes and strengthen the system along the way, continuously.

Moreover, an effective red team operation would not end after it has found a weakness in the system. That is just the beginning. The team's next role is to provide remediation assistance and re-testing. Also, more importantly, keep doing this as long as necessary.

There may be significant work involved in comprehending the findings, their impact, likelihood, criticality and detectability. Furthermore, carrying out suggested remediations, retraining of the machine with new data, etc before the blue team says it is ready for the next round of testing is needed.

The whole process of the red team finding weaknesses and the blue team fixing these has to be an ongoing process with regular checks and balances. Avoid the temptation to do it once for the sake of it. Instead, make sure that you do it regularly and consistently. Doing so will help you to maintain a watch on the risk score of each aspect and monitor how you are progressing with the already-established risk mitigation plan. Your target for each risk item in the list should be to reduce its risk score to near-zero. **EFY**